

Computational Analysis of Human Insulin Protein Sequence

Overview

This report presents a computational analysis of the human insulin protein sequence (Homo sapiens) using bioinformatics tools. The study involves sequence retrieval from NCBI, structural and functional annotation, and similarity analysis using BLAST.

Data Retrieval from NCBI

The insulin protein sequence was retrieved from the NCBI database. The protein analyzed is insulin from Homo sapiens with accession number AAA59172.1 and a sequence length of 110 amino acids.

The sequence represents preproinsulin, which undergoes post-translational modifications to form active insulin.

Sequence Features Analysis

The protein consists of several functional regions including a signal peptide (residues 1–24), which directs the protein to the secretory pathway, and a mature peptide (residues 25–110) that forms biologically active insulin.

The C-peptide region connects the A and B chains during protein folding. The protein belongs to the insulin-like growth factor (IGF) family and plays a crucial role in glucose metabolism and growth regulation.

BLAST Analysis

BLAST was used to identify similar protein sequences. The results show high similarity with insulin sequences from primates such as chimpanzees, bonobos, and macaques.

The query coverage is approximately 100 percent with percent identity ranging between 97 to 99 percent. The E-values are extremely low, indicating highly significant matches.

Evolutionary Insights

Insulin is highly conserved across species, especially among primates. Minor variations exist, but the overall sequence similarity highlights its evolutionary importance and essential biological function.

Biological Significance

Insulin plays a critical role in regulating blood glucose levels and carbohydrate metabolism. Dysfunction of insulin production or activity leads to conditions such as diabetes mellitus.

The structural conservation of insulin ensures proper receptor binding and functional stability.

Conclusion

This analysis demonstrates that human insulin is a highly conserved protein with well-defined functional regions. BLAST results confirm strong evolutionary relationships across species. Bioinformatics tools such as NCBI and BLAST are essential for sequence analysis, functional prediction, and evolutionary studies.

Tools Used

NCBI Protein Database and BLAST.